

XRPL Validator — Daily Engineering Report • 2026-08-29

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **13 commits** (xrpl-validator 13)

Engineering changes

- **feat(xrpl-node): XRPL_REPLAY_TIMING=1 — per-ledger apply/hash phase receipts** (c9dc9c2 , xrpl-validator) The Stage 4 perf gate: raw native-apply cost with no RPC in the path, on its own TIMING line so gate greps of REPLAY lines stay stable. Verdict over 280 ledgers / 29,464 txs: apply avg 20.6ms, median 19.9ms, p95 37.6ms, max 63.9ms — 62x headroom at worst against the 4s ledger cadence. The 11.6s avg encode+fold is the replay's own full re-fold of 14M entries, not the engine (the live validator hashes incrementally in ...
- **feat(xrpl-node): Stage 4 Phase A — native-engine shadow leg (XRPL_NATIVE_SHADOW=1)** (2d66eb0 , xrpl-validator) The native Rust transaction engine now runs beside the FFI (libxrpl C++) leg on every live ledger, dormant until armed.
- **fix(xrpl-node): shadow leg stamps threading — first-light lesson** (02465de , xrpl-validator) #106628655, the first ledger the native engine ever shadowed live: 175 byte diffs, 5 missing keys, ZERO TER mismatches — the exact signature of PreviousTxnID/PreviousTxnLgrSeq never being stamped. The replay applies stamp_threading after every tx; the shadow loop now does the identical call. The mirror's canonical reconciliation did its job: nothing compounded.
- **feat(xrpl-node): shadow receipts name their classes — typed missing keys, stale-mirror audits, reconcile-leak verifier** (fcb7b3a , xrpl-validator) Typed missing keys (decode the FFI bytes for LedgerEntryType — the first histogram read 'deleted'x41, the fingerprint of a deletion leak). TER mismatches audit the mirror against the meta's own PreviousFields and print which field of which class went stale (first catch: AccountRoot.OwnerCount mirror=4 truth=3, staleness born ledgers before the symptom). Post-reconcile verifier re-reads every overlay key and ...
- **fix(xrpl-ledger): funds-exhausting fill sizes from the gross-cap chain, not the rem_gets accumulator** (c53b5af , xrpl-validator) Finding 30, first blood of the Stage 4 live shadow (#106629200 9264210055, OfferCreate RLUSD->XRP, five fills, unrated, underfunded): slices 1-4 byte-identical; the funds-exhausting slice 5 paid the me_sub chain's remainder (...443496) where rippled pays the stored line's remainder (...443500) — the maker's owed line must close to EXACT ZERO and ours landed -4e-15. The rem_gets me_sub chain and the line's STAmount adds ...
- **fix(xrpl-ledger): equal-withdraw fraction ceils, asset products floor** (bf4a729 , xrpl-validator) Findings 31+32 — one rule pinned from both sides of a single tfWithdrawAll (#106629211 7ED17A0F, ARMY/DROP two-IOU pool): the DROP payout demanded a larger result than frac-nearest-then-floor gives, the ARMY payout

a smaller one than any product-up gives, and the unique satisfying pair is `st_divide_up` (exact 16-digit ceil, new helper) for the fraction with `mul_directed` floor for both products. The dust calibrator ...

- **fix(xrpl-ledger): AMMVote — default-zero omission and AuctionSlot DiscountedFee** (`0ac8d7e` , xrpl-validator) Finding 33, two gaps in the two-day-old `applyVote` port, one specimen (`#106629215 AC9BA657`): (a) a `VoteEntry` voting fee ZERO serializes WITHOUT its `TradingFee` (`SoeDefault` omission — ours was 3 bytes long per zero-fee entry), same rule applied to the AMM root `TradingFee`; (b) the `AuctionSlot`'s `DiscountedFee` follows every vote — `fee/10`, set when nonzero, REMOVED when zero (`AMMVote.cpp:208-218`) — ours left the stale ...
- **feat(xrpl-node): shadow dumps the engine's-eye tx on first per-ledger TER mismatch** (`d57e19b` , xrpl-validator) Defect B (live-only TER storms the offline replay refutes): the first mismatching tx per ledger now logs its parsed fields exactly as `build_txfields` delivered them — if the live inputs differ from the offline world's in any byte, the receipt names it; if they match, the divergence is environmental to the process and the dump proves that.
- **fix(xrpl-ledger): equal-withdraw fraction ceil gates on fixAMMv1_3** (`c26156a` , xrpl-validator) The census caught the over-rotation: `#105880685` (pre-amendment) lands one ULP HIGH under an unconditional ceil while `#106629215` (post) demands it — an amendment boundary, exactly where rippled splits (`rules().enabled(fixAMMv1_3)`, `AMMHelpers getRoundedAsset`). New `amendment_enabled()` reads the `Amendments` singleton straight off sandbox state, so every world — probe hydration, full-state replay, the live mirror — ...
- **fix(xrpl-ledger): equal-withdraw fraction gates on WITHDRAW MODE, not amendment** (`4185a4a` , xrpl-validator) The census's second look ended the amendment theory: `fixAMMv1_3` predates BOTH calibrators (active at `#105880684` per the `Amendments` singleton on the wire). The true discriminator is the withdraw mode: `tfWithdrawAll` ceils the fraction (`#106629211`, both pool sides pinning it uniquely), `tfLPToken` partials keep the legacy nearest fraction (`#105880685` one ULP HIGH under an unconditional ceil). `withdraw_all` threads through ...
- **fix(xrpl-node): mirror speaks the engine's dialect — hexify addresses on every decode** (`02e0bbb` , xrpl-validator) Defect B, solved by its own field dump in one receipt: the engine's state convention is HEX account fields (every offline world — snapshot loader, probe hydration — runs `hexify_addresses` after decode), and hex-only parsers like `check.rs parse_account_id` read r-addresses as ABSENT. The live mirror inserted raw decoder output, so `CheckCancel`'s creator compare failed as `tecNO_PERMISSION` on ledgers the replay proves ...
- **fix(xrpl-node): shadow hydration — clear-before-fill and live-edge gating** (`9badf09` , xrpl-validator) v3's death, from its own log: hydration's 2-minute stall raises lag past the blob-fetch gate (`lag<=25`), the overlay vanishes for the catch-up ledgers, the next overlay ledger gaps past the mirror, the gap triggers re-hydration, and the loop repeats — while `hydrate()` INSERTED into the existing 19.8M-object map without clearing, so the second pass ~doubled RSS and the OOM killer ended the third (silent SIGKILL, no ...

Docs and operations

- **Merge branch 'loop/differential-harness'** (`06c8658` , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Fractional NFT Co-Ownership via Vault Shares** (nft) Add a tfNFTokenVault flag to VaultCreate (SingleAssetVault) letting the funding deposit be a single locked NFTTokenID instead of a fungible amount; the
- **Scoped, Expiring Multisign Session Entries** (accounts) Add three optional SFields to each SignerEntry inside an ExpandedSignerList — sfTxTypeMask (bitmap of allowed tt codes), sfMaxAmount, and sfExpiration
- **Consensus-Enforced Reserve Backing for MPT Issuers** (audit) Add sfReserveAccount (AccountID) and sfMinReserveBps SFields to MPTokenIssuance, set at MPTokenIssuanceCreate and immutable unless the issuance carries

Appendix — commits

Time	Repo	Commit	Subject
01:07	xrpl-validator	c9dc9c2	feat(xrpl-node): XRPL_REPLAY_TIMING=1 — per-ledger apply/hash phase receipts
15:03	xrpl-validator	06c8658	Merge branch 'loop/differential-harness'
15:20	xrpl-validator	2d66eb0	feat(xrpl-node): Stage 4 Phase A — native-engine shadow leg (XRPL_NATIVE_SHADOW=1)
15:31	xrpl-validator	02465de	fix(xrpl-node): shadow leg stamps threading — first-light lesson
15:53	xrpl-validator	fc7b73a	feat(xrpl-node): shadow receipts name their classes — typed missing keys, stale-mirror audits, reconcile-leak verifier
17:35	xrpl-validator	c53b5af	fix(xrpl-ledger): funds-exhausting fill sizes from the gross-cap chain, not the rem_gets accumulator
18:59	xrpl-validator	bf4a729	fix(xrpl-ledger): equal-withdraw fraction ceils, asset products floor
19:34	xrpl-validator	0ac8d7e	fix(xrpl-ledger): AMMVote — default-zero omission and AuctionSlot DiscountedFee
19:35	xrpl-validator	d57e19b	feat(xrpl-node): shadow dumps the engine's-eye tx on first per-ledger TER mismatch
20:53	xrpl-validator	c26156a	fix(xrpl-ledger): equal-withdraw fraction ceil gates on fixAMMv1_3
22:03	xrpl-validator	4185a4a	fix(xrpl-ledger): equal-withdraw fraction gates on WITHDRAW MODE, not amendment
23:08	xrpl-validator	02e0bbb	fix(xrpl-node): mirror speaks the engine's dialect — hexify addresses on every decode
23:20	xrpl-validator	9badf09	fix(xrpl-node): shadow hydration — clear-before-fill and live-edge gating